FORECASTING THE S&P 500 INDEX VOLATILITY

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ABSTRACT

This study introduces a new pre-differencing transformation for the ARIMA model for forecasting S&P 500 index volatility. The out of sample forecasting performance of the ARIMA model using the new pre-differencing transformation is compared with the out of sample forecasting performance of the mean reversion model and the GARCH model. The ARIMA model using the new pre-differencing transformation introduced in this study is found to be superior to both the mean reversion model and the GARCH model in forecasting monthly S&P 500 index volatility for the forecast comparison periods used in this study.

I. INTRODUCTION

Forecasting stock market volatility is important for several reasons. Stock market volatility forecasts are important in the timing of investment decisions. In addition, the expected stock market volatility is an important input in portfolio selection models and dynamic diversification techniques. Risk averse investors may choose to adjust their portfolios by shifting a portion of their assets away from assets whose volatilities are expected to increase in the future.

Volatility forecasting also plays an important role in the pricing of contingent claims such as stock options. The future volatility of the underlying asset return is an explicit argu-

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ment in the various option-pricing models. To derive the price of a option according to these models, a forecast of the future volatility of the underlying asset return is needed for the remaining life of the option.

Another area where stock market volatility forecasts are important is in dynamic portfolio insurance strategies. The market volatility forecast is an important input in dynamic portfolio insurance strategies that implement a put replication program with futures contracts. Hill, Jain, and Wood (1988) show that volatility over the period that the insurance program is in place is a critical factor in estimating the costs of portfolio insurance.

Thus, forecasting stock market volatility is important for portfolio managers and investors. More accurate stock market volatility forecasting models may help improve the performance of existing option pricing models, portfolio selection models and market timing models. Better stock market volatility forecasting models can also help portfolio managers using dynamic portfolio insurance strategies to select the appropriate delta for their insurance horizon.

The autoregressive conditional heteroscedastic (ARCH) model introduced by Engle (1982) and the generalized autoregressive conditional heteroscedastic (GARCH) model developed by Bollerslev (1986), which is an extension of the ARCH model to allow for a more flexible lag structure, have been used frequently in recent studies to model volatility. Akgiray (1989) compares several time series models in terms of their ability to forecast monthly stock market volatility using the Center of Research in Security Prices (CRSP) value- weighted and equal-weighted indexes and finds the GARCH model provides the best forecast of stock market volatility. Randolph and Najand (1991) in comparing the GARCH model against the mean reversion model find the mean reversion model to be superior to the GARCH model in forecasting the monthly volatility of S&P 500 stock index futures for the three-year sample comparison period 1986-1988 used in the study. In a related study, Randolph (1991) shows that the mean reversion model outperforms the GARCH model in forecasting monthly volatilities of the S&P 500 index for the sample comparison period from May 1986 to September 1987.

The contributions of this study are as follows. This study shows, using more recent data and a larger forecast comparison period from 8/17/88 to 1/31/94, that the mean reversion model is not able to outperform the GARCH model in forecasting monthly S&P 500 index volatility. This study also finds that when the more volatile sub-period from 8/17/88 to 12/3/90 is used as the forecast comparison period, the mean reversion model performed slightly better than the GARCH model in forecasting monthly S&P 500 index volatility in terms of the mean average percentage error (MAPE), mean absolute error (MAE), and root mean square error (RMSE). Finally, this study introduces a new pre-differencing transformed ARIMA model which is superior to both the GARCH model and the mean reversion model in forecasting monthly S&P 500 index volatility for both the smaller sub-sample forecast comparison period from 8/17/88 to 12/3/90 and the entire forecast comparison period from 8/17/88 to 1/31/94 used in this study.

II. DATA

Daily data from April 1983 to January 1994 of the S&P 500 cash index from the Futures Industry Institute is used in this study. The sample data set of daily S&P 500 cash index prices is divided into two parts. The first part is from 4/21/82 to 8/16/88. The second part is from 8/17/88 to 1/31/94. The second part of the data set serves as the test or comparison period in which the out of sample forecasts from the models are compared. The first part of the data set is reserved for estimating the initial parameters of the models.

Since there are approximately 20 trading days in a calendar month, the actual volatility for each month in this study is calculated from the log of the price relatives of the S&P cash prices using a period of 20 days to calculate each volatility. Specifically, the actual average daily volatility for month p is calculated as:

$$\sigma_p = \left(\frac{\sum_{i=1}^{n} (R_p - \mu_p)^2}{n-1} \right)^{1/2} \qquad (1)$$

where n is equal to 20, $R_p = \log(P_t/P_{t-1})$ is defined as the natural logarithm of price relatives of the S&P 500 cash index prices. P_t is the S&P 500 cash index price and P_{t-1} is its price the previous day.

Following Randolph and Najand (1991) the average daily standard deviation for a particular month is also referred to in this study as a monthly volatility.

It should be noted that the applicability of the use of the standard deviation as a measure of volatility is dependent on the specific functional form of the distribution of stock returns. Gray and French (1990), for example, find distributions other than the normal to be better descriptors of the distribution of the S&P 500 and S&P 100 returns. For such non-normal distributions, they suggest that the scale parameter for the distribution may be a better measure of volatility. However, equation (1) is still widely used as an estimator of stock return volatility especially in the option pricing literature. Randolph (1991), Randolph and Najand (1991), and Canina and Figlewski (1993) are examples of recent studies that use the standard deviation (equation (1)) as an estimator of volatility.

For option pricing, it is important to have an accurate forecast of the future standard deviation of returns over the remaining life of an option to be able to value the option accurately. For example, the standard deviation of returns is an explicit argument in the popular option pricing models. In addition, both the GARCH model and the mean reversion model were used in previous studies to forecast the standard deviation of returns as in equation (1). In order to be able to compare the model developed in this study with models used in previous studies and because of its importance in options pricing, the standard deviation of returns is, thus, used in this study as an estimator of volatility. Technically speaking, the models discussed in this study are models for forecasting the future standard deviation of returns as in equation (1). However, in as much as the standard deviation of returns are an estimator of volatility, we can loosely refer to these models as 'volatility' forecasting models.

After the initial parameters of the models are estimated or set using the data in the reserved estimation period, out of sample forecast of the average standard deviation of the daily returns for the next 20 days is made using the models. This forecast is also referred to as the monthly volatility forecast in this study since there are approximately 20 trading days in a calendar month. Once a period is forecast, it then joins the information set. Successive periods are forecast for a total of 70 out of sample volatility forecasts.

III. MODELS

A. Mean Reversion Model

The benchmark mean reversion model used in this study to forecast stock index return volatility is the same one as the model used by Randolph (1991) to forecast stock index return volatility. The mean reversion model has been used earlier in a study by Cox, Ingersoll, and Ross (1985) to model short-term rate of interest. The mean reversion model has also been used in simulating results for the stochastic volatility option pricing model. Wiggins (1987) and Hull and White (1988) are examples of studies which use the mean reversion model in this manner.

The mean reversion model assumes that the stock futures index X and the instantaneous standard deviation σ satisfy the following stochastic differential equations:

$$dX = \mu \, Xdt + \sigma X dW \tag{2}$$

$$d\sigma = \delta\sigma dt + \theta\sigma dZ \tag{3}$$

where dW and dZ are Wiener processes with correlation r, μ, θ, and δ are parameters. δ is the drift parameter of the volatility process and is assumed to follow a mean reversion process:

$$\delta = a + b\,\sigma \tag{4}$$

where a and b are constants and $a > 0$ and $b < 0$. θ is the standard deviation of σ, or the volatility of the volatility.

The mean reversion model uses the following equation to forecast volatility:

$$E[\sigma_{p+1}] = e^{bT}[\sigma_p - MRL] + MRL \tag{5}$$

where $E[\sigma_{p+1}]$ is the expected volatility for period $p + 1$ given information available at period p and MRL is the mean reversion level.

The selection of the mean reversion level provides the target toward which the volatility returns after stochastic departures. Randolph (1991) suggests that the MRL be set equal to the long term average of the volatility series under consideration. In this study, the MRL is set equal to .009565, the long term average daily standard deviation of returns on the S&P 500 for the sample estimation period from 4/21/82 to 8/16/88.

The forecasts of the mean reversion model are produced using the prior month's volatility as the volatility input σ_p.

Following Randolph (1991), a time factor of 20 days is used to make the monthly mean reversion forecasts. Operationally, this means that setting T in equation (5) equal to 20 implying that the input volatility σ_p reverts toward the MRL for 20 days.

The mean reversion rate, b, gives the half life for the mean reversion process. The mean reversion rate, b, in equation (5) is estimated using non-linear least squares using the data in the model estimation period from 4/21/82 to 8/16/88 and is updated as each new observation becomes available in a rolling fashion.[1]

B. GARCH Model

The generalized autoregressive conditional heteroscedastic (GARCH) model was developed by Bollerslev (1986) and is an extension of the autoregressive conditional heteroscedastic (ARCH) model introduced by Engle (1982) to allow for a more flexible lag structure. The GARCH model has been used frequently in recent studies to model volatility. This study uses the GARCH(1,1) model outlined in Akgiray (1989) and used in both Randolph (1991) and Randolph and Najand (1991) for forecasting monthly volatility. Specifically, given daily stock index return data, the GARCH(1,1) model can be described as follows:

$$R_t | \Omega_{t-1} \sim F(\mu_t, V_t) \tag{6a}$$

$$\mu_t = \Phi_0 + \Phi_1 R_{t-1} \tag{6b}$$

$$v_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}{}^2 + \beta_1 v_{t-1} \tag{6c}$$

$$\varepsilon_t = R_t - \Phi_0 - \Phi_1 R_{t-1} \tag{6d}$$

where $\alpha_0 > 0$, $\alpha_1 >= 0$, $\beta_1 >= 0$ and $F(\mu_t, v_t)$ is the conditional distribution of the variable, with mean μ_t, and variance v_t. The information set available at time t is given by Ω_{t-1}.

Using the GARCH(1,1) model, a forecast for the variance over a 20 day period can be calculated using the following:

$$v_m = \sum_{t=1}^{20} \left(\frac{(1 - (\Phi_1)^t)}{(1 - \Phi_1)} \right)^2 \left(A^{20-t} v_t + \sum_{j=0}^{19-t} \alpha_0 A^j \right) \tag{7}$$

where $A = \alpha_1 + \beta_1$ and v_t is given by equation (6c).

Following Randolph and Najand (1991), this study transforms the forecast monthly variance into the forecast daily standard deviation V_{MF} for the 20 day period, specifically, $V_{MF} = \sqrt{(v_m/20)}$.

C. Pre-differencing Transformed ARIMA Model

In order to use the ARIMA model with pre-differencing transformation to forecast, we first transform the σ_p time series into a time series of probabilities $P(T)_p$ using the following equation:

$$P(T)_p = \Phi\left(\frac{-\alpha + \left(\bar{\mu} - \frac{1}{2}\bar{\sigma}^2\right)T}{\sigma_p\sqrt{T}}\right) + \exp\left(\frac{2\left(\bar{\mu} - \frac{1}{2}\bar{\sigma}^2\right)\alpha}{\sigma_p^2}\right)\Phi\left(\frac{-\left(\bar{\mu} - -\frac{1}{2}\bar{\sigma}^2\right)T - \alpha}{\sigma_p\sqrt{T}}\right) \tag{8}$$

$\Phi\,[\cdot]$ is the cumulative normal distribution function. σ_p is the volatility for period p. T represents a time horizon and is set equal to 20 corresponding to the number of trading days in each period used in this study. $\bar{\mu}$ is the long run average daily log return of the S&P 500 index. $\bar{\sigma}$ is the long run average daily standard deviation of daily returns of the S&P 500 index.[2] α is a distance parameter. $P(T)_p$ is the probability measure for month p.

The intuition behind using equation (8) to transform the original time series is that with an appropriate parameter value for α, it stabilizes the variance of the time series, thereby producing a transformed time series which is more suitable for ARIMA modeling. Equation (8) gives us the probability that a Brownian motion particle will travel a distance α sometime within a time period T.[3] In other words, if we define $\mu' = \left(\bar{\mu} - \frac{1}{2}\bar{\sigma}^2\right)$, then if we assume that the log of the S&P 500 index during any particular month p follows a Brownian motion process with drift parameter $\mu'$and average daily volatility equal to σ_p, then equation (8) can be alternatively interpreted as the probability that the log S&P 500 index will travel a distance α sometime within a time period T.

Notice that different values for α will produce different transformed series $P(T)_p$. Setting the value of α too large would cause the $P(T)_p$ series to bunch up near the value of 0. On the other hand, setting the value of α too low would cause the $P(T)_p$ series to bunch up near the value of 1. A $P(T)_p$ series bunched up near 0 and a $P(T)_p$ series bunched up near 1 is not as forecastable as a $P(T)_p$ series with a mean of approximately .5. The optimal value for the parameter α can be found by minimizing the following equation with respect to α using the data in the reserved estimation period:

$$\min_{\alpha}\left(.5 - \left(\Phi\left(\frac{-\alpha + \left(\bar{\mu} - \frac{1}{2}\bar{\sigma}^2\right)T}{\sigma_p\sqrt{T}}\right) + \exp\left(\frac{2\left(\mu - \frac{1}{2}\bar{\sigma}^2\right)\alpha}{\sigma_p^2}\right)\Phi\left(\frac{-\left(\mu - \frac{1}{2}\bar{\sigma}^2\right)T - \alpha}{\sigma_p\sqrt{T}}\right)\right)\right)^2 \tag{9}$$

The value of α produced by equation (9) makes the mean of the transformed $P(T)_p$ series reasonably close to .5.[4]

Univariate ARIMA is then used to forecast $P(T)_{p+1}$. In this study, the constructed $P(T)_p$ series is found to be reasonably described by an ARIMA(0,1,2) process. Once a value for $P(T)_{P+1}$ is forecasted, it is plugged back into equation (8) and used to back out a value for σ_{p+1}, the forecast for next month's volatility. This process is repeated for each period in the comparison period generating a total of 70 out of sample monthly volatility forecasts using the pre-differencing transformed ARIMA model.

D. Naive Model

It is also of interest to compare the results of the naive model with the other models. The naive model simply assumes that the forecast volatility is the same as the current volatility (the most recently observable volatility). The model is:

$$\hat{\sigma}_{p+1} = \sigma_p \tag{10}$$

where $\hat{\sigma}_{p+1}$ is the volatility forecast made at the end of period p for period $p + 1$.

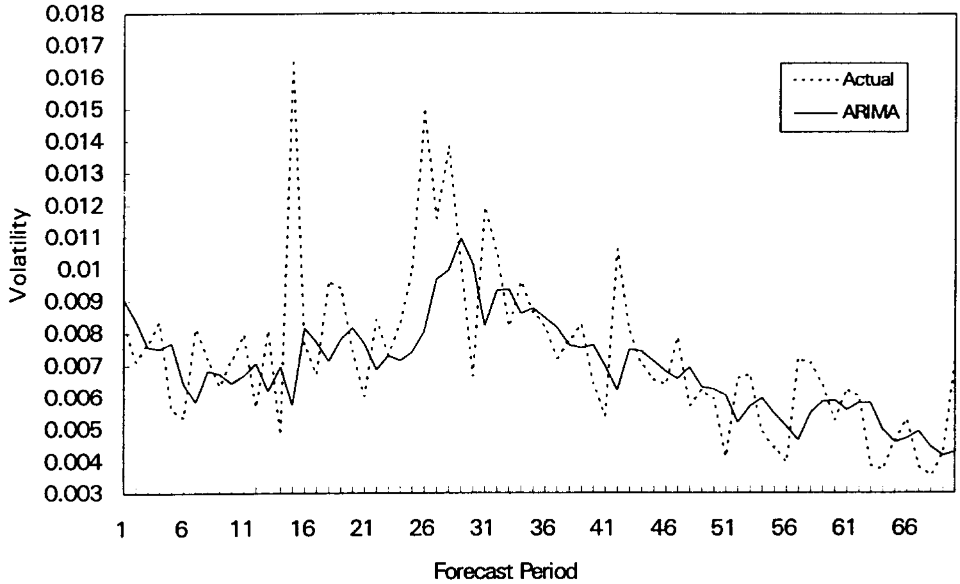
IV. RESULTS

A. Summary Statistics

Figures 1–3 display the actual average daily standard deviation for each forecast period and the forecasts produced by the pre-differencing transformed ARIMA model, the GARCH model, and the mean reversion model respectively. These figures show that both the GARCH model and the mean reversion model consistently produce volatility forecasts that are too high toward the later part of the forecast comparison period. Toward the later part of the forecast comparison period, the actual S&P 500 index volatility gradually declined. However, the mean reversion model is not able to track this gradual decline and forecasts from this model are consistently higher than the actual S&P 500 index volatility during this period. The GARCH model did a slightly better job than the mean reversion model of tracking this decline in actual S&P 500 index volatility in the later part of the forecast comparison period, but it still tends to produce volatility forecasts that are higher than the actual index volatility. Unlike both the mean reversion model and the GARCH model, the pre-differencing transformed ARIMA model is able to track the decline in the actual index volatility in the later part of the forecast comparison period.

Table 1 lists the mean absolute percentage error (MAPE), mean absolute error (MAE), root mean square error (RMSE), and mean percentage error (MPE) for the out of sample forecasts produced by the four models in the study. The top part of Table 1 shows the results for the sub-period from 8/17/88 to 12/3/90 calculated using a total of 30 out of sample forecasts. The middle part of Table 1 shows the results for the sub-period from 12/4/90 to 1/31/94 calculated using a total of 40 out of sample forecasts. The bottom of Table 1 shows the results for the entire forecast comparison sample period from 8/17/88 to 1/31/94 calculated using a total of 70 out of sample forecasts.

Table 1 shows that the pre-differencing transformed ARIMA model outperforms the mean reversion model, the GARCH model, and the naive model for both the second forecast comparison sub-period and entire forecast comparison period in terms of MAPE, MAE, RMSE, and MPE. For the first forecast comparison sub-period when the actual stock index volatility was more volatile, both the GARCH and the mean reversion model performed better than the naive model in terms of MAE and RMSE. In terms of MAPE, however, the mean reversion model performed better than the naive model, but the



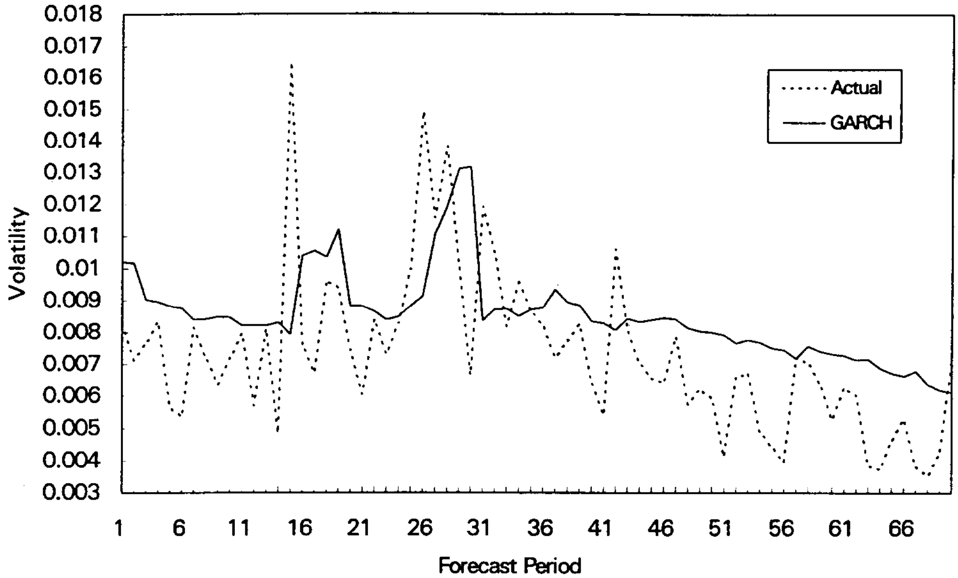
Note: Each forecast period in Figure 1 represents 20 trading days. The actual volatility for each period is calculated from the log of the price relatives of the S&P cas prices using a period of 20 days to calculate each volatility.

Figure 1. Pre-differencing Transformed ARIMA Model Forecasts of the S&P 500 Cash Index Volatility (8/17/89 to 1/31/94)

GARCH model slightly underperformed the naive model. Furthermore, for the first forecast comparison sub-period when the actual stock index volatility was more volatile, the mean reversion model performed slightly better than the GARCH model in terms of all of the given forecast performance comparison summary statistics. These results seem to suggest that the mean reversion model may forecast stock index volatility better than the GARCH model during relatively more volatile time periods and when the average S&P 500 index volatility is close to the long run mean reversion level but may be inferior to the GARCH model when the stock index volatility is more stable and when the stock index volatility drifts away from the long run mean reversion level.

For the second forecast comparison sub-period when the actual stock index volatility gradually declined, both the GARCH and the mean reversion model underperformed the naive model in terms of the given forecast performance comparison summary statistics. For the entire forecast comparison period from 8/17/88 to 1/31/94 the GARCH model outperformed the naive model with respect to the RMSE. However, with respect to MAPE, MAE, and MPE, the naive model outperformed both the GARCH and the mean reversion model.

The MPE summary statistic gives an indication whether a forecasting method is biased (consistently forecasting low or high). If the forecasting approach is unbiased, the MPE summary statistic will produce a percentage that is close to zero. If the result is a large negative percentage, the forecasting method is consistently overestimating. If the result is a



Note: Each forecast period in Figure 2 represents 20 trading days. The actual volatility for each period is calculated from the log of the price relatives of the S&P cash prices using a period of 20 days to calculate each volatility.

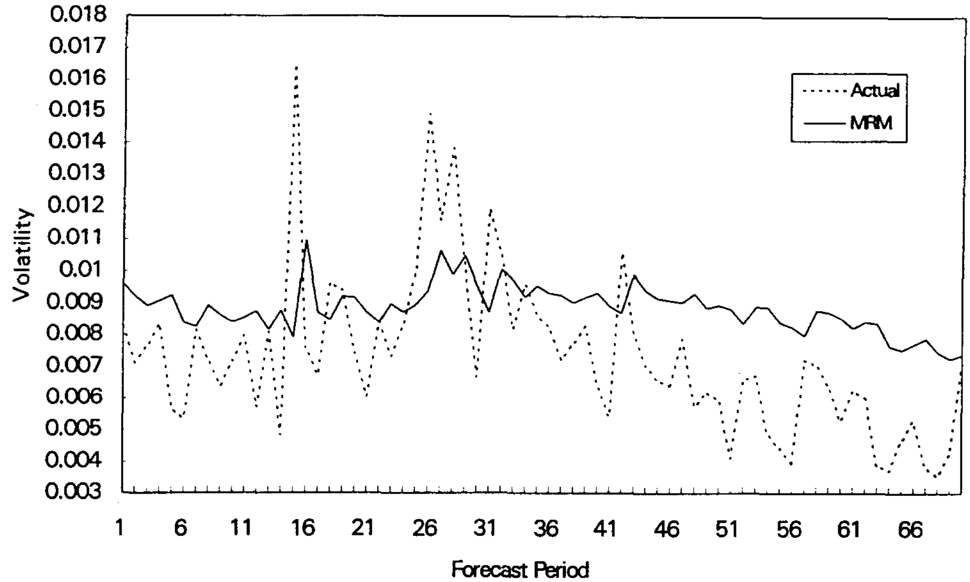
Figure 2. GARCH Model Forecasts of the S&P 500 Cash Index Volatility (8/17/89 to 1/31/94)

large positive percentage, the forecasting method is consistently underestimating. As shown in Table 1, the small MPEs of both the pre-differencing transformed ARIMA model and the naive model suggests that these two models are not biased. On the other hand, the MPEs of both the GARCH and the mean reversion model are notability larger than the MPEs of the pre-differencing transformed ARIMA model and naive model indicating that these two models produced forecasts that are biased in that these two models seem to be consistently overestimating during the forecast comparison period. For the first and more volatile comparison sub-period, the MPE of the mean reversion model is less than the MPE of the GARCH model. However, for the second forecast comparison period when the S&P 500 index volatility gradually declined, the MPE of the GARCH model is less than the MPE of the mean reversion model.

B. Regression Tests

The regression tests used in this section to test the predictive ability of each one of the alternative volatility forecasting models are the same as those used in Shastri and Tandon (1986), Canina and Figlewski (1993), and Jorion (1995). A regression of the following type is used to compare the four models:

$$\mathit{actual\ volatility}_p = \alpha + \beta X_p + \mu_p \tag{11}$$



Note: Each forecast period in Figure 3 represents 20 trading days. The actual volatility foe each period is calculated from the log of the price relatives of the S&P cash prices using a period of 20 days to calculate each volatility.

Figure 3. Mean Reversion Model Forecasts of the S&P 500 Cash Index Volatility (8/17/89 to 1/31/94)

$$for\ X_p = GARCH,\ ARIMA, MRM,\ Naive$$

X_p is the forecasted volatility for period p from the GARCH model, pre- differencing transformed ARIMA model, the mean reversion model, and the Naive model. μ_p is the regression residual. The R^2 would give an indication of the explanatory power of the alternative models. A better forecast should produce a higher R^2.

In addition, if we regress the actual volatility for period p on two model forecasts say $F_{1,p}$ and $F_{2,p}$;

$$\underset{volatility_p}{actual} = \alpha + \beta_1 F_{1,p} + \beta_2 F_{2,p} + \mu_p \tag{12}$$

if neither model 1 nor model 2 contains any information useful for forecasting, the estimates of β_1 and β_2 should both be zero. If both models contain information useful for forecasting then both β_1 and β_2 should be non zero. If both models contain information, but the information contained in, say, model 2 is completely contained in model 1 and model 1 contains further relevant information as well, then β_1 but not β_2 should be non zero.

The results of the regression tests as shown in equation (11) are reported in Table 2. These results indicate that for the forecast comparison period used in this study the performance rankings of the forecasting models are (from the best performing to the worst performing): 1) pre-differencing transformed ARIMA model 2) GARCH model 3) Naive model 4) mean reversion model. It is interesting to note that both the pre-differencing transformed ARIMA

Table 1. Summary Statistics for Out of Sample Model Forecasts of the S&P 500 Cash Index Volatility

	GARCH	*MRM*	*ARIMA*	*NAIVE*
Forecast Comparison Sample Period: (8/17/88 to 12/3/90)				
MAPE	0.285322	0.259201	0.208284	0.278536
MAE	0.002236	0.002048	0.001908	0.002405
RMSE	0.002961	0.002737	0.002840	0.003407
MPE	−0.205266	−0.159310	0.024338	−0.074778
Forecast Comparison Sample Period: (12/4/90 to 1/31/94)				
MAPE	0.330101	0.461655	0.173761	0.198941
MAE	0.001775	0.002461	0.001090	0.001323
RMSE	0.002050	0.002736	0.001430	0.001790
MPE	−0.282228	−0.435148	−0.026889	−0.031736
Forecast Comparison Sample Period: (8/17/88 to 1/31/94)				
MAPE	0.310910	0.374889	0.188557	0.233053
MAE	0.001972	0.002284	0.001441	0.001787
RMSE	0.002482	0.002736	0.002151	0.002609
MPE	−0.249244	−0.316932	−0.004934	−0.050182

Notes: Table 1 lists the mean absolute percentage error (MAPE), mean absolute error (MAE), root mean square error (RMSE), and mean percentage error (MPE) for the out of sample forecasts of 20 day standard deviations of the S&P 500 Cash Index produced by the GARCH model, mean reversion model, pre-differencing transformed ARIMA model, and the naive model.

$$\text{MAPE} = \frac{1}{N}\sum_{p=1}^{N}\left|\frac{actual_p - forecast_p}{actual_p}\right|$$

$$\text{MAE} = \frac{1}{N}\sum_{p=1}^{N}\left|actual_p - forecast_p\right|$$

$$\text{RMSE} = \left[\frac{1}{N}\sum_{p=1}^{N}(actual_p - forecast_p)\right]^{1/2}$$

$$\text{MPE} = \frac{1}{N}\sum_{p=1}^{N}\frac{(actual_p - forecast_p)}{actual_p}$$

is the actual volatility for period p and $forecast_p$ is the forecasted volatility for period p.
N is the number of periods in the forecast comparison period of interest.

model and the GARCH model outperformed the naive model in the regression tests whereas the mean reversion model was outperformed by the naive model in these tests.

The results of the regression tests as shown in equation (12) are reported in Table 3. Table 3 shows the results of comparing the information contained the pre-differencing transformed ARIMA model (the best performing model from the regression tests in Table 2 with the information contained in the GARCH model, the mean reversion model, and the naive model respectively. The results in Table 3 shows that the pre-differencing transformed ARIMA model is informationally superior to all of the other models analyzed in this study for the forecast comparison period used in this study.

Table 2. Regression Tests of the Predictive Ability of Model Forecasts

$$\text{actual volatility}_p = \alpha + \beta X_p + \mu_p$$

X_p	α	*SE*	β	*SE*	R^2
ARIMA	.00117	.00123	.88691	.17383	.276
GARCH	−.00003	.00166	.86582	.19236	.229
Naive	.00404	.00083	.44687	.10771	.201
MRM	−.00506	.00341	1.40249	.38412	.163

Note: The table reports regression results for equation (11) for the forecast comparison period from 8/17/88 to 1/31/94 using a total of 70 observations.

V. CONCLUSIONS

This study introduces a new pre-differencing transformation for the ARIMA model for forecasting S&P 500 index volatility. The out of sample forecasting performance of this model is compared with the out of sample forecasting performance of the mean reversion model, the GARCH model, and the naive model. The mean reversion model was shown by Randolph (1991) to outperform the GARCH model in forecasting the S&P 500 index volatility for the 18 months comparison period used in their study from May 1986 to September 1987. This study shows, using more recent data and a larger forecast sample period from 8/17/88 to 1/31/94, that the mean reversion model is no longer able to outperform the GARCH model in forecasting monthly S&P 500 index volatility. This study also finds that when the more volatile sub-period from 8/17/88 to 12/3/90 is used as the forecast comparison period, the mean reversion model performed slightly better than the GARCH model in forecasting monthly S&P 500 index volatility in terms of the MAPE, MAE, and RMSE. Finally, the results of the regression tests showed that the new pre-differencing transformed ARIMA model introduced in this study is informationally superior to the mean reversion model, the GARCH model, and the naive model in forecasting monthly S&P 500

Table 3. Regression Tests of the Pre-differencing Transformed ARIMA Model Versus the Others

Other Model	*Constant*	*ARIMA*	*Other*	R^2
GARCH	.00062	.73733	.18664	.279
	(.381)	(2.158)	(.509)	
MRM	.00498	1.15545	−.64119	.285
	(1.146)	(3.380)	(−.912)	
Naive	.00118	.88453	.00164	.276
	(.875)	(2.633)	(.008)	

Note: The table reports regression results for equation (12) for the forecast comparison period from 8/17/88 to 1/31/94 using a total of 70 observations. t-ratios are in parentheses.

index volatility for the forecast comparison period from 8/17/88 to 1/31/94 used in this study.

NOTES

1. For forecasting the S&P 500 index volatility, Randolph (1991) recommends using a long run estimate of b set equal to a constant value of approximately -.06. Preliminary testing, however, showed that estimating and updating b using the non-linear least squares technique described in this study gave better results and thus is used in this study.
2. The long run average daily log return and the long run average daily standard deviation of the S&P 500 index was shown by Gray and French to be approximately .000543 and .00879 respectively.
3. See Cox and Miller (1965) for detailed derivation and discussion of why the formula given in equation (8) gives us the probability that a Brownian motion particle will travel a distance α sometime within a time period T.
4. The optimal α was found to be approximately equal to .02944 in this study.

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